

Synthesis and Characterization of Novel Biodegradable Di- and Tri-Block Copolymers Based on Ethylene Carbonate Polymer as Hydrophobic Segment

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ABSTRACT: Synthesis of novel amphiphilic biodegradable block copolymers based on ethylene carbonate is reported in this study. Polyethylene glycol monomethyl ether (MeO-PEO) and polyethylene glycol (PEG) of varying molar masses are used as macro-initiator for ring-opening polymerization of ethylene carbonate in the presence of sodium stannate trihydrate as a heterogeneous transesterification catalyst. Earlier elution of block copolymer from macro-initiator in size exclusion chromatography (SEC) indicated the successful synthesis of the block copolymers. Ratios of both types of blocks are varied systematically. Liquid chromatography at critical conditions is used for the analysis of the non-critical individual blocks, and if there

are any critical segments that are not attached to the non-critical block. To the best of our knowledge, this is the first report on the synthesis of ethylene carbonate-based amphiphilic block copolymers. Chromatographic critical conditions of the ethylene carbonate polymer are also reported for the first time. © 2017 Wiley Periodicals, Inc. *J. Polym. Sci., Part A: Polym. Chem.* **2017**, *55*, 1887–1893

KEYWORDS: amphiphilic block copolymers; critical adsorption point of poly(ethylene ether-carbonate); liquid chromatography at critical conditions; poly(ethylene ether-carbonate)

INTRODUCTION Search of novel and efficient drug delivery systems is one of the hot research topics of current science. An inherent part of drug delivery and other medical-related formulations are biocompatible, biodegradable, or bioresorbable polymers.^{1,2} Moderate rate of bio-degradation and non-toxic degradation products of aliphatic polycarbonates make them a captivating candidate for the above-mentioned applications.^{3,4}

Ring-opening of cyclic carbonates is the most suitable method for the synthesis of fairly high molar mass polycarbonates.^{5,6} Block copolymers containing polycarbonate segments have been reported by several scientists in order to induce peculiar slow degradation and other mechanical properties.^{7–9} Amphiphilic biodegradable block copolymers are very important materials. Their amphiphilic behaviour, mechanical, and physical properties can be manipulated by varying relative block lengths, molar mass, and chemical composition of the blocks, for required applications.^{10,11}

Polyethylene oxide is the most widely used hydrophilic segment of synthetic biodegradable block copolymers. It is non-toxic, biocompatible and is easily excreted from body through urinary system.^{12–14} The block copolymers of PEO

with polycarbonates are alluring candidate for medical-related applications due to their peculiar biodegradability, slow degradation rates, and non-acid degradation products. Amphiphilic behaviour of the block copolymers can be tuned by changing the type of the carbonate segment and/or the relative lengths and composition of both blocks. There are several literature reports on the synthesis of polycarbonate-based amphiphilic block copolymers.^{15–17} In above-mentioned reports, poly(trimethylene carbonate), poly(2,2-dimethylene carbonate), poly(5-methyl-5-allyloxycarbonyl-propylene carbonate), and poly(5-allyloxy-1,3-dioxane-2-one) have been used as hydrophobic block. Hydrophobicity of the polycarbonate block can be tailored by inducing different pendant groups such as OH, NH₂, and COOH on the polymer chain, that can be further used for grafting.^{17,18} In none of the above-mentioned literature, polymer synthesized from ethylene carbonate monomer is used as hydrophobic segment. In this study, we would like to report first-time synthesis of amphiphilic block copolymers containing ethylene carbonate-based polymer as hydrophobic block.

ROP of EC is associated with formation of carbon dioxide during polymerization that results in ether linkages in the

polymer chains alongside targeted carbonate linkages.^{19–21} Cationic ROP is less prone to expulsion of carbon dioxide compared to anionic ROP of EC.^{19,20,22,23} In one of our recent study, we reported on comprehensive structural elucidation of the polymer synthesized by ROP of ethylene carbonate in presence of sodium stannate trihydrate as transesterification catalyst.¹⁹ It has been shown that the ratio of carbonate to ether linkages is slightly less than one (less than 50% carbonates linkages). For the sake of consistency, poly(ethylene ether carbonate), PEEC, shall be used throughout the text for the polymer segment formed by polymerization of ethylene carbonate, since it is more representative of the structure. Sodium stannate trihydrate catalyzed ROP of ethylene carbonate is conducted for the synthesis of block copolymers that is initiated by polyethylene glycols of varying molar masses (hydrophilic block). Polyethylene glycol monomethyl ethers of varying molar masses ranging from 550 to 5000 g/mol are used as macro-initiator for synthesis of di-block copolymers. The same range of molar mass of poly(ethylene glycol)s is used as macro-initiator for synthesis of tri-block copolymers with PEEC as outer and PEG as inner block. The ratios of both blocks are varied systematically to maneuver systematic changes in the amphiphilic behaviour. Synthesized block copolymers are analyzed by different modes of liquid chromatography of polymers including size exclusion chromatography and liquid chromatography at critical conditions. Augmentation in the molar mass after addition of PEEC block to the macro-initiator is monitored by size exclusion chromatography. Molar mass of individual block, and absence or presence of unwanted homopolymers of both types is determined by liquid chromatography at critical conditions. Liquid chromatography at critical conditions has been used for analysis of the individual block lengths of the block copolymers,^{24–35} and for end-group analysis of telechelics and functional polymers.^{29–32,36–40} To the best of our knowledge, the novel block copolymers having PEEC as hydrophobic block are first time reported in this study. Critical adsorption point of PEEC is also reported for the first time and is used for the analysis of the novel biodegradable block copolymers.

HPLC OF POLYMERS

Currently, three liquid chromatographic methods are mostly utilized to characterize polymers, namely, size exclusion chromatography (SEC), Interaction liquid chromatography (iLC) or liquid adsorption chromatography (LAC), and liquid chromatography under critical conditions (LCCC).

SEC is the simplest, and most widely used technique for characterization of synthetic and natural polymers.^{41,42} Separations by SEC are entropy controlled and depend on the hydrodynamic volume of the analyte in dilute solution. It is not possible to separate polymers with similar hydrodynamic volume in solution but varying in architecture, chemical composition, or topology. Polymers elute in order of decreasing hydrodynamic volume. SEC can be used as a first test for

the augmentation in the hydrodynamic size of the polymers in solution that is related to the relative molar mass by external calibration, however, SEC cannot provide any information about chemical composition or if the polymer sample is a mixture of two types of homopolymers or all polymer segments are attached to each other making block copolymer.

For complex polymeric systems, interaction modes of liquid chromatography reveal more information because separation could be achieved with regard to the interaction strength of the analyte with the stationary phase. These modes are particularly useful for determination of unwanted homopolymers in the copolymer samples. Interaction chromatographic modes can be isocratic or gradient including both solvent and temperature gradients.^{29–32,43–46}

Larger polymers are strongly adsorbed, and consequently are held longer, while smaller polymers are only slightly adsorbed and elute earlier. Interaction liquid chromatography (iLC) has better resolution than SEC, however, it is not as universal as SEC. For a specific polymer, the strength of the enthalpic interaction has to be controlled to accomplish a reproducible separation. Selection of a suitable stationary phase, mobile phase, and gradient (either solvent or temperature) is not universal and often requires extensive experimentation. The recovery of the polymers, and in turn their quantitation is also influenced by events such as precipitation of the polymers in the injector, breakthrough peaks, and irreversible adsorption of large polymers.^{43,47,48} At the intersection of exclusion (entropy controlled) and interaction modes (enthalpy controlled) of liquid chromatography of polymers, there is a special combination of mobile phase, stationary phase, temperature and analyte, where all the polymers elute at the same elution volume independent of their molar mass. This special point is called critical adsorption point (CAP) and this mode of liquid chromatography of polymers is termed as liquid chromatography under critical conditions (LCCC).^{29–32}

LCCC effectively allows separation of complex polymers with regard to distributions other than molar mass such as functionality,^{29–32,36–40} and the other block.^{24–35} In LCCC, exclusion and adsorption effects counterbalance each other, and critical homopolymers are transported along the column become 'chromatographically invisible', which means that their elution is no more dependent on the molar mass. Thus, in the case of a block copolymer, one block can be eluted under critical conditions while the other block can elute in either exclusion^{28–32,36,49} or adsorption mode.^{28–36} This allows separation of one of the parent homopolymers from the block copolymer, and separation of block copolymer is achieved with regard to the length of the non-critical block. The limitations of LCCC are the same as discussed for interaction modes of LC of polymers such as limited recovery, irreversible adsorption etc. Another limitation of LCCC-SEC is decreased accuracy of the molar mass distribution of non-critical block with the increasing length of the critical block, nonetheless, it provides reasonable estimates.²⁵

TABLE 1 Synthesis Protocols of Di- and Tri-Block Copolymers

S. No.	Macro-Initiator : Monomer (by Weight)	Macro-Initiator (PEG/MeO-PEO) (g)	Monomer (Ethylene Carbonate) (g)	Catalyst (Sodium Stannate Trihydrate) (g)
1	3:1	3.0	1.0	0.005
2	1:1	2.0	2.0	0.01
3	1:3	1.0	3.0	0.015

EXPERIMENTAL

Material

Ethylene carbonate (Daejung, Korea), diethylene glycol (Scharlau, Spain), Sodium Stannate trihydrate (Sigma Aldrich, USA) were used as received. PEGs and MeO-PEOs of varying molar masses (PEG₄₀₀, PEG_{1K}, PEG_{2K}, PEG_{4K}, MeO-PEO₅₅₀, MeO-PEO₇₅₀, MeO-PEO_{2K}, and MeO-PEO_{5K}) were purchased from Sigma-Aldrich (Germany) and were dried by azeotropic distillation with toluene before use.

HPLC grade solvents acetonitrile, acetone (Fisher Scientific, USA) and CHCl₃ (RCI Labscan, Thailand) were used as received. Double deionized water (Millipore, Progard 2, China) was used.

Polymerizations

PEEC homopolymers were synthesized according to the procedure described in our recent publication.¹⁹ For synthesis of di- and tri-block copolymers polyethylene glycol monomethyl ethers (MeO-PEO) and polyethylene glycols (PEG) of varying molar masses are used as macro-initiator, respectively. As a typical procedure, calculated amounts of dried macro-initiator, monomer, and catalyst (0.5 wt % of monomer) were added in an oven-dried flask, and is subjected to three evacuation and nitrogen purging cycles. The polymerizations were carried out at 175 °C under inert atmosphere. The polymerization times were varied according to the targeted length of the PEEC block. MeO-PEO and PEG of varying molar masses (PEG₄₀₀, PEG_{1K}, PEG_{2K}, PEG_{4K}, MeO-PEO₅₅₀, MeO-PEO₇₅₀, MeO-PEO_{2K}, and MeO-PEO_{5K}) are used for synthesis of di- and tri-block copolymers, respectively. Table 1 summarizes the synthesis protocols of di- and tri-block copolymers.

Liquid Chromatography

An Agilent SEC instrument (infinity 1200 series) equipped with an isocratic pump, manual injector, column oven and a RI detector was used for SEC analysis. A set of three PLgel columns (300 mm × 7.5 mm, 5 μm, Mixed D) connected in front with a PLgel guard column (50 mm × 7.5 mm) is used. The column oven was maintained at 30 °C. Chloroform (HPLC grade) was used as an eluent at a flow rate of 1.0 mL/min. The SEC system was calibrated with narrow molar mass polystyrene standards (ranging from 165 g/mol to 3 × 10⁶ g/mol) by Polymer Standard Services (Germany). Therefore, all the molar mass values obtained by SEC analysis are relative to linear PS. The data were processed by Agilent OpenLAB ChemStation. Sample concentrations were kept

between 3.0 and 5.0 mg/mL, and 100 μL of sample solution was injected.

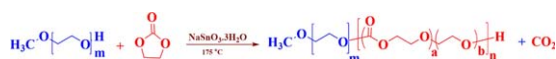
Liquid chromatographic analyses at critical conditions were performed at an Agilent technologies chromatograph (infinity 1200 series). The system is equipped with a binary pump, a column oven, an auto-sampler, and an evaporative light scattering detector. The column oven was maintained at 30 °C. The temperature of ELSD was set at 40 °C and nitrogen gas was used as carrier gas at a pressure of 50 psi. The mobile phase was delivered at 0.5 mL/min. Sample concentrations were kept between 3.0 and 10.0 mg/mL, and 20 μL of sample solution was injected. The data were processed by Agilent Open LAB ChemStation for LC system. Following columns were used in this study.

- ZORBAX ODS, pore size 70 Å, particle size 5 μm, and column dimensions 250 × 4.6 mm (Phenomenex, USA)
- Ultisil diol column, pore size 120 Å, particle size 5 μm, and column dimensions 250 × 4.6 mm (Welch, China)

HPLC grade acetone, acetonitrile, and double deionized water were used for preparation of mobile phases. Solvents were mixed by volume and vacuum degassed.

RESULTS AND DISCUSSION

The polymerization of ethylene carbonate by sodium stannate trihydrate as a heterogeneous transesterification catalyst, and the relative content of the carbonate to ether linkages have been described in our recent publication, through comprehensive structural analysis by advanced NMR techniques.¹⁹ The coordination ring-opening polymerization of ethylene carbonate can be initiated by hydroxyl end-group. The hydroxyl end-group of polyethylene glycols (macro-initiator) is used as initiating moiety for polymerization of ethylene carbonate to synthesize di- and tri-block copolymers (see Scheme 1). There can be expulsion of carbon dioxide during polymerization that result in ether linkages in the polymer block based on ethylene carbonate. Nonetheless, the ratio of ether and carbonate linkages remains around one under all conditions. Therefore, both blocks shall have quite different properties from each other and products can show peculiar self-assembly behaviour. The specific hydrophilic-lipophilic balance of these biodegradable amphiphilic block copolymers can be exploited for novel applications. The hydrophilic-lipophilic balance can be further tailored by varying block lengths and compositions of the both segments. In this study, lengths of both blocks are varied



SCHEME 1 Polymerization of ethylene carbonate with MeO-PEO as initiator. [Color figure can be viewed at wileyonlinelibrary.com]

systematically by using macro-initiator of different lengths and altered ratios of the monomer to macro-initiator.

Size exclusion chromatograms of some representative samples are shown in Figure 1. Earlier elution of block copolymers from macro-initiator suggests the increase in the hydrodynamic size, and in turn the molar mass of the polymer species. Elution volume decreased with increasing amount of ethylene carbonate monomer in the reaction mixture. SEC can be a good first test of the reaction products for molar mass propagation, however, it cannot provide any information regarding presence of both types of unwanted homopolymers in the sample due to its limited resolution. Reliable method for determination of the absence or presence of parent homopolymers in the samples could be different modes of interaction chromatography. Liquid chromatography at critical conditions is the method of choice for such analysis.

Liquid chromatography at critical conditions has been used for analysis of polymers with regard to functionality,^{29–32,36–40} or other block either in interaction^{28–36} or size exclusion regime.^{28–32,36,49} Chromatographic critical conditions for numerous polymers under several sets of mobile phase-stationary phase and temperature have been reported in literature.^{28–32} The critical adsorption point (CAP) for polycarbonates (PC) are never reported due to several possible reasons such as absence of commercially available homopolymer samples of varying molar masses and tedious procedure for establishment critical adsorption point.^{29–32}

First step for any LCCC method development is the selection of suitable stationary phase that allows differentiation between two segments of the block copolymer. Critical conditions of comparatively less polar segment on RP column shall allow the exclusion behaviour of the polar segment. In this particular case, at critical adsorption point of PEEC on RP column, PEGs should elute in exclusion regime. Therefore, elution behaviour of varying molar masses of PEEC on RP column in a ternary mobile phase mixture is evaluated on a RP column. Mobile phase used is aqueous acetonitrile with 1.0% triethylamine (TEA) as an additive. Addition of TEA is necessary to avoid any undesired polar interaction of free silanol groups with ethylene oxide or ether units.²⁸ As shown in Figure 2(A), pure acetonitrile with 1.0% TEA is a strong solvent for PEEC and the elution volume decreased with increase in the molar mass (exclusion regime). Addition of water in the mobile phase shifts the selectivity of the PEEC more towards the stationary phase. In mobile phase composition acetonitrile-water-TEA:59-40-1, PEEC elute in the adsorption mode, the elution volume increased with the

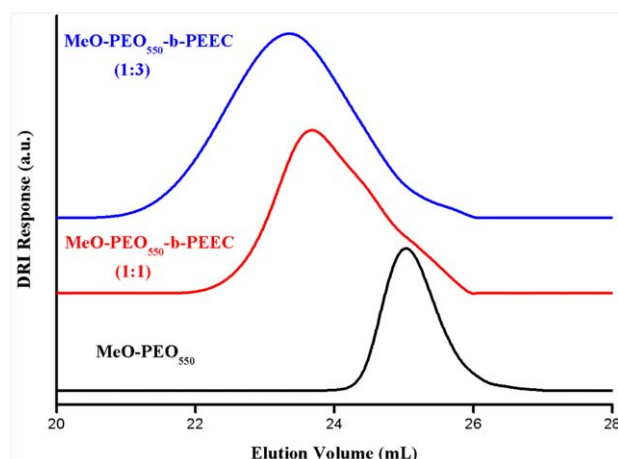


FIGURE 1 Molar mass propagation as shown by size exclusion chromatography for a typical di-block copolymer with different monomer to macro-initiator ratios; A set of three PLgel columns (300 mm × 7.5 mm, 5 μm, Mixed D), mobile phase: CHCl₃, flow rate: 1.0 mL/min, RI detector; System calibrated by linear polystyrene standards (ranging from 165 g/mol to 3 × 10⁶ g/mol); see Table 1 for synthesis protocols. [Color figure can be viewed at wileyonlinelibrary.com]

molar mass, Figure 2(B). The chromatographic critical behaviour was observed in mobile phase composition acetonitrile-water-TEA:64-35-1, v/v. With this mobile phase composition, PEEC with molar masses ranging from 500 to 5000 g/mol eluted at the same elution volume independent of their molar mass. Although, polymer of ethylene carbonate is

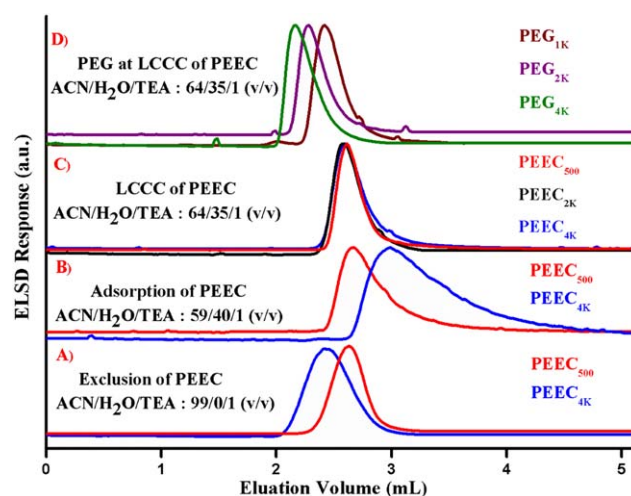


FIGURE 2 Elution behaviour of PEEC in different mobile phase compositions on a ZORBAX ODS RP 70 Å, 5 μm, 250 mm × 4.6 mm column, detector: ELS, column temperature: 30 °C; Elutograms of PEEC of varying molar masses with different compositions of mobile phases; acetonitrile-water-TEA, A) 99-0-1 (v/v %) (Exclusion mode), B) 59-40-1 (v/v %) (Adsorption mode), C) 64-35-1 (v/v %) (Critical adsorption point of PEEC); D) Elution behaviour of PEGs at CAP of PEEC – acetonitrile-water-TEA (v/v %):64-35-1 (v/v %). [Color figure can be viewed at wileyonlinelibrary.com]

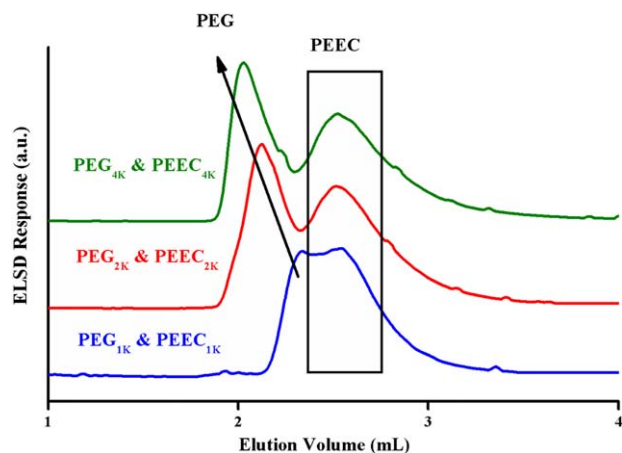


FIGURE 3 Elugrams of blends of PEEC and PEG at CAP of PEEC on a ZORBAX ODS RP 70 Å, 5 μ m, 250 mm $\times$ 4.6 mm column, detector: ELS, column temperature: 30 $^{\circ}$ C; mobile phase acetonitrile-water-TEA (v/v %):64-35-1 (v/v %). [Color figure can be viewed at wileyonlinelibrary.com]

composed of ether linkages along with targeted carbonate linkages, the comprehensive NMR analysis showed that the polymer is not random but statistical copolymer.¹⁹ The successful establishment of critical conditions of the polymer endorse the NMR results. Statistical polymers behave like a homopolymer in chromatography, equivalent to homopolymer of some effective segment that is repeated throughout the chain.^{50,51} In this particular case, EC-EE-EE might behave as an effective repeat unit, Figure 2(C). To the best of our knowledge, this is the first report of critical adsorption point of PEEC. The PEGs elute in order of decreasing molar mass at critical conditions of PEEC on RP column, typical size exclusion behaviour [Fig. 2(D)].

Figure 3 depicts the elugrams of the blends of similar molar mass samples of PEEC and PEG at CAP of PEEC. PEEC eluted independent of their molar mass at their critical conditions, however, PEGs are excluded from the pores of the stationary phase and eluted in order of decreasing molar mass.

Figure 4 demonstrates the elution behaviour of di- and tri-block copolymers of ethylene carbonate with varying molar mass of PEO macro-initiator at critical conditions of PEEC. Initial reaction mixture of the products shown in Figure 4, contain equal amounts of macro-initiator and ethylene carbonate monomer; therefore, similar molar mass of both blocks is expected. The block copolymers are separated according to molar mass distribution of noncritical PEO. As can be noticed, there is no maximum observed at the elution volume of critical PEEC for most of the products (2.56 mL). A shoulder extending to the elution volume of critical PEEC for the product initiated with PEG_{4K} suggest the presence of some PEEC in the sample that are not attached to the macro-initiator. To confirm the accuracy of the developed method, one of the tri-block copolymers is spiked with PEEC. The difference between spiked and un-spiked samples clearly showed that if there would be any significant amount of

PEEC in the sample that is not attached to the macro-initiator (PEG), it should have been separated as a separate peak at its critical adsorption point.

Critical conditions of PEEC revealed information about molar mass distribution of noncritical PEO block, and presence or absence of PEEC segments that are not attached to the macro-initiator.

For determination of molar mass distribution of PEEC block, and presence or absence of PEG in the samples, analysis at critical adsorption point of PEG is required. Critical conditions of PEG have been reported using diverse mobile phases on several stationary phase of varying polarity.^{24,52,53}

In this study, we chose critical adsorption point of PEG on a NP diol column.^{24,52} In a binary aqueous mixture of 70 vol% acetone, PEG_{4K} eluted later than PEG₅₀₀, showing that PEG adsorbed on the stationary phase under these conditions, Figure 5(A). The elution order is reversed in 60% acetone [see, Fig. 5(B)]. The critical adsorption point of PEG should be between these two mobile phase compositions. PEG of different molar masses eluted at same elution volume irrespective of their molar mass in mobile phase containing 65% acetone, Figure 5(C). Because of low polarity of PEEC compared with PEG, they are expected to elute in SEC order at the CAP of PEG on NP column. Figure 5(D) depicts that the PEEC eluted in order of decreasing molar mass at critical adsorption point of PEG on NP column, as expected.

Figure 6 presents the elugrams of the blends of similar molar masses of PEEC and PEG at CAP of PEG. In contrary to RP, the molar mass independent elution of PEG is accompanied by early elution of PEEC in order of decreasing molar mass, a typical SEC behaviour.

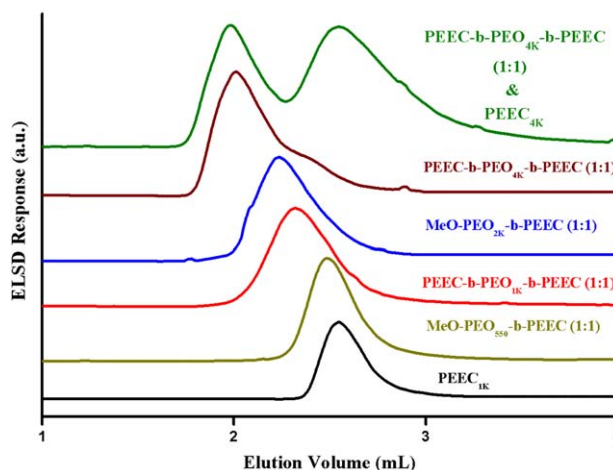


FIGURE 4 Elugrams of di- and tri-block copolymers of PEO with PEEC at CAP of PEEC on a ZORBAX ODS RP 70 Å, 5 μ m, 250 mm $\times$ 4.6 mm column, detector: ELS, column temperature: 30 $^{\circ}$ C; mobile phase acetonitrile-water-TEA (v/v %):64-35-1 (v/v %); see Table 1 for synthesis protocols. [Color figure can be viewed at wileyonlinelibrary.com]

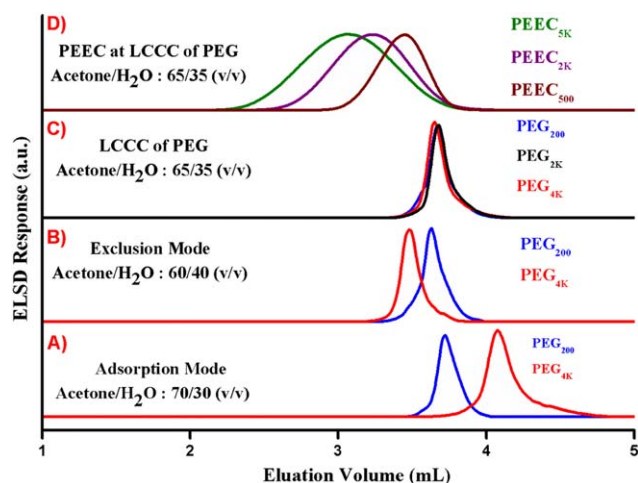


FIGURE 5 Elution behaviour of PEG in different mobile phase compositions on a Ultisil diol column, 120 Å, 5 μ m, 250 mm $\times$ 4.6 mm (Welch, China), detector: ELS, column temperature: 30 $^{\circ}$ C; Elugrams of PEG of varying molar masses with different mobile phases, A) 70% acetone in water (exclusion mode), B) 60% acetone in water (adsorption mode), C) 65% acetone in water (critical adsorption point of PEG); D) Elution behaviour of PEEC at CAP of PEG – 65% acetone in water. [Color figure can be viewed at [wileyonlinelibrary.com](#)]

Di- and tri-block copolymers synthesized in current study, should show similar elution behaviour as that of PEEC under these chromatographic conditions. At the same time, the method should be able to separate if there is any unreacted macro-initiator in the sample. As a typical case, a block copolymer with increasing length of PEEC block is shown in Figure 7. Earlier elution of the products with more ethylene carbonate content in the initial reaction mixture, indicates the greater length of the non-critical block. It is also noticed that there is some unreacted macro-initiator for the product with comparatively less initial ethylene carbonate content (25%). The content of unreacted macro-initiator decreased

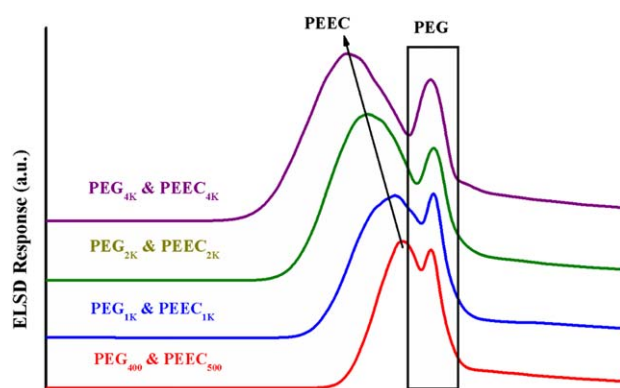


FIGURE 6 Elugrams of blends of PEEC and PEG of similar molar masses at CAP of PEG on a NP Ultisil diol column, 120 Å, 5 μ m, 250 mm $\times$ 4.6 mm (Welch, China), detector: ELS, column temperature: 30 $^{\circ}$ C; mobile phase: 65% acetone in water. [Color figure can be viewed at [wileyonlinelibrary.com](#)]

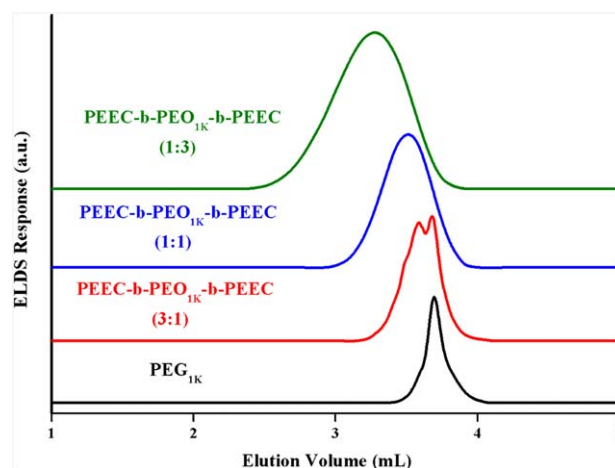


FIGURE 7 Elugrams of tri-block copolymers of PEO_{1K} with PEEC having different ratios of macro-initiator to ethylene carbonate monomer, at CAP of PEG on a NP Ultisil diol column, 120 Å, 5 μ m, 250 mm $\times$ 4.6 mm (Welch, China), detector: ELS, column temperature: 30 $^{\circ}$ C; mobile phase: 65% acetone in water; see Table 1 for synthesis protocols. [Color figure can be viewed at [wileyonlinelibrary.com](#)]

with increasing amount of ethylene carbonate monomer in the reaction mixture and all the macro-initiator reacted for the product with 75% ethylene carbonate monomer in the initial reaction mixture.

The amphiphilic behaviour of these novel block copolymers, effect of lengths of both blocks and their relative amount on the critical micelle concentration, the size and shape of the micelles shall be subject of our next publication of this series.

CONCLUSION

Synthesis of di- and tri-block amphiphiles based on poly(ethylene oxide) as hydrophilic block and poly(ethylene ether-carbonate) segment as hydrophobic part is reported for the first time. Block copolymers are characterized by liquid chromatography at critical conditions. Critical conditions of PEG have been reported by several authors. However, the critical adsorption point of hydrophobic segment PEEC is never reported in literature. The hydrophobic segment also contains ether linkages alongside targeted carbonate linkages. The statistical nature of the hydrophobic segment as revealed by NMR analysis¹⁹ is endorsed by chromatography. Statistical nature of the segment allowed for the establishment of critical adsorption point. The analysis of the individual molar mass of non-critical block, and presence of free hydrophilic or hydrophobic critical segments is monitored by liquid chromatography at critical conditions. It has been shown that in case of lesser amount of ethylene carbonate monomer in the sample, some of the macro-initiator remains unreacted. However, with increased amount of ethylene carbonate in the reaction mixture all the macro-initiators are initiated, making a part of block copolymer.

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